

Do girls actually run the world? Where in the world would you most likely be a minister as a woman?

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Introduction

Women make up about half of the world's population, yet they are a minority in political decision-making. Ministers lead government departments, so the share of women among them shows how much women take part in the decisions that affect everyone. Equal participation of women in political leadership is also an official goal of the United Nations (Sustainable Development Goal 5), which makes this share a measurable indicator of progress toward gender equality in politics.[1]

This project asks:

1. What percentage of ministers in governments worldwide are women and how did this percentage change between 2005 and 2024.

2. How do world regions compare in the year with the highest world average?

The code was also made to estimate on how long would it take to reach gender parity if the current pace continues to show how far away from said sustainable goal our world is.

Data sample

The dataset [2], "Share of women in ministerial positions", comes from Our World in Data and is based on data from the Inter-Parliamentary Union and UN Women, distributed through the World Bank. The file has 2438 values for 193 countries in 13 years between 2005 and 2024, with no missing values. Twelve aggregate rows (World, income groups, European Union, World Bank regions) were removed from the country analysis. The six World Bank region rows and the world row are used for the region comparison.

Methods

The analysis was written in Python (numpy, matplotlib). For each year the share is averaged over all countries, with the standard error (standard deviation divided by the square root of the number of countries) as the statistical uncertainty. Outliers are removed in each year with the interquartile-range rule: values more than 1.5 times the interquartile range (how spread out the middle half of the data is) below the lower quartile or above the upper quartile. This removes 43 values (1.8%). A straight line is fitted to the yearly averages and extrapolated to 50%.

Results and discussion

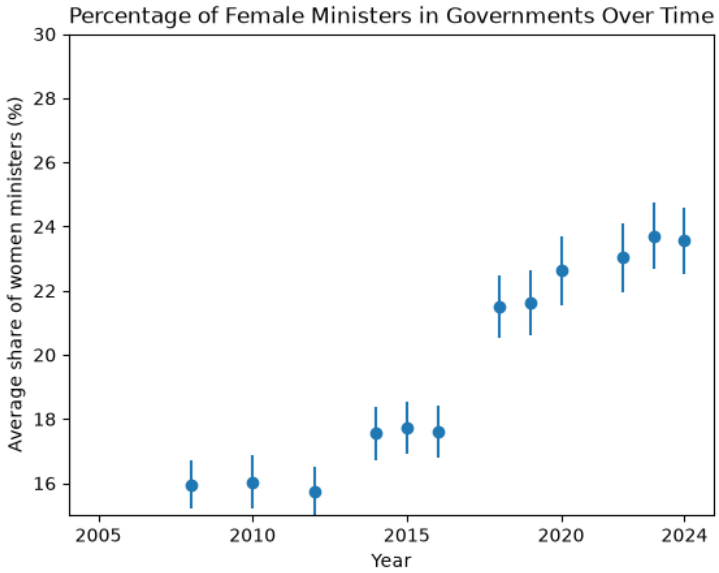


Figure 1: Percentage of female ministers in governments over time (2005–2024).

Figure 1 shows that the average share of women among ministers rose from 13.1 ± 0.7 % in 2005 to 23.6 ± 1.0 % in 2024. The rise is not steady: the average jumped from 17.6 % in 2016 to 21.5 % in 2018, then stayed near 23 % after 2022. Countries with at least 50 % women ministers grew from 2 in 2005 to 15 in 2023, then fell to 9 in 2024.

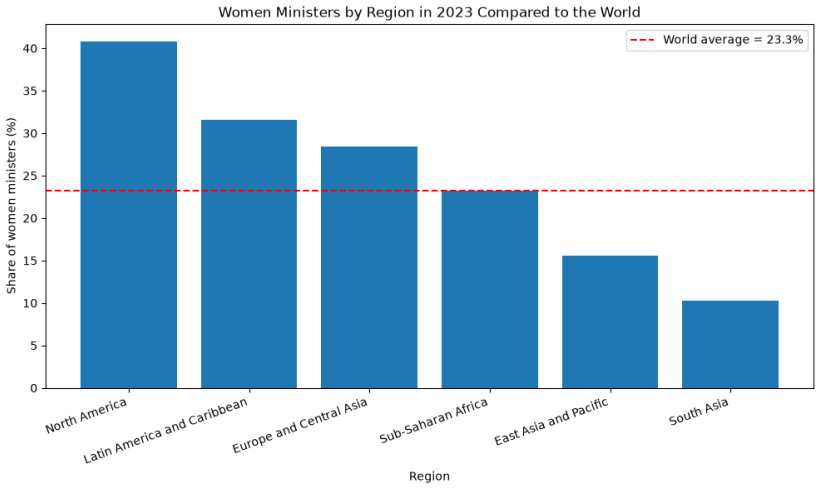


Figure 2: Women ministers by World Bank region in 2023.

The world average is highest in 2023 (23.3 %), so Figure 2 (right) compares World Bank regions in that year. North America is highest (40.8 %, +17.6 points above the world), followed by Latin America and the Caribbean (31.5 %) and Europe and Central Asia (28.4 %). Sub-Saharan Africa (23.2 %) matches the world average, while East Asia and Pacific (15.6 %) and South Asia (10.2 %, -13.0 points) are below it.

About one in four ministers is a woman. If the linear trend continues, 50 % would be reached in about 45 years (around 2069), but the recent plateau makes this uncertain. Ministers are only a proxy for political power, and removing outliers (43 values) steepens the trend (0.59 instead of 0.51 points per year).

Conclusion

This project analyzed the share of women in ministerial positions worldwide, using Python for data cleaning, statistics and visualization.

To summarise the key findings:

- The average share of women ministers grew from 13.1 ± 0.7 % (2005) to 23.6 ± 1.0 % (2024), by 0.59 ± 0.05 percentage points per year.
- In 2023, regional shares ranged from 10.2 % (South Asia) to 40.8 % (North America), compared with a world average of 23.3 %.
- At the current pace, parity (50 %) would be reached in about 45 years, but the trend is not steady.

References

[1] United Nations. Sustainable Development Goal 5: Gender equality. https://sdgs.un.org/goals/goal5

[2] IPU via World Bank; UN Women via World Bank – processed by Our World in Data. “Share of women in ministerial positions”. Available at: https://archive.ourworldindata.org/20260817-180151/grapher/share-of-women-in-ministerial-positions.html.